THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** compass	geometry tool
2	**(d)** all sides are equal	special rectangle
3	**(c)** perpendicular bisector of AB	equal distance
4	**(a)** length and breadth	two sides
5	**(a)** square	equal sides
6	**(a)** all four angles are right angles	definition
7	**(c)** 2	mid-lines
8	**(b)** diameter	through the centre
9	**(c)** 40 cm	$2(12+8)$
10	**(a)** 62 cm	$2(20+11)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite sides equal and parallel; all angles are 90° (diagonals equal).	1+1
12	A square has all sides equal; a rectangle needs only opposite sides equal.	1+1
13	They are equal and bisect each other at right angles.	1+1
14	A line that cuts a segment into two equal parts at 90°.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw 6 cm base (1); set square for 90° at both ends and mark 4 cm (1); join the top points (1).	1+1+1
16	Draw 5 cm base (1); raise two 90° lines of 5 cm (1); join the ends (1).	1+1+1
17	Equal (1); bisect each other (1); divide rectangle into congruent triangles (1).	1+1+1
	OR: Draw arcs of equal radius from both ends above and below (1.5); join the intersections to cut at the midpoint (1.5).	1½+1½
18	A square has all rectangle properties plus equal sides (1.5); a rectangle may have unequal sides (1.5).	1½+1½
	OR: Correct circle with labels (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Properties (2). (ii) Construction (2). (iii) Both measure about 8.6 cm (1).	2+2+1
	OR: (i) Properties (2). (ii) Construction (2). (iii) 90° (1).	2+2+1
20	(i) Correct (1). (ii) Steps and drawing (2). (iii) Measure the distances (2).	1+2+2
	OR: (i) Two right-angled triangles (1). (ii) Yes, congruent (1). (iii) 4,800 cm² (1). (iv) Diagonal divides the rectangle in two equal parts (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** radius	fixed opening
2	**(a)** equal in length	property
3	**(c)** cuts it at its midpoint at 90°	definition
4	**(a)** a ruler and set square	sliding method
5	**(c)** two congruent right triangles	property
6	**(d)** compass or divider	measure
7	**(b)** 4	mid-lines and diagonals
8	**(b)** perpendicular bisector of AB	equal distance from A and B
9	**(d)** 28 cm	$2(7+7)$
10	**(b)** 16 cm	$2(5+3)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ruler, compass, protractor, set squares, divider.	1+1
12	Distance from centre to any point on the circle.	1+1
13	10 cm (equal diagonals).	1+1
14	They meet at the centre dividing each into equal halves.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	All sides equal (1); all angles 90° (1); diagonals equal and perpendicular (1).	1+1+1
16	Area of the rectangle 108 cm² (1); each triangle 54 cm² (2).	1+2
17	Draw AB (1); with compass arcs from A and B (1); join the intersections (1).	1+1+1
	OR: OA = 7 cm (1.5); BD = 14 cm (1.5).	1½+1½
18	Rectangle 60 × 40 cm; area 2,400 cm² (1.5); perimeter 200 cm (1.5).	1½+1½
	OR: Equal arcs on both sides of the point and then two arcs above (1.5); join to get the perpendicular (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Explanation (2). (ii) Rectangle: equal, bisect; square: equal, bisect at 90° (3).	2+3
	OR: (i) Circle (1). (ii) Two chords (1). (iii) Bisectors (2). (iv) Meet at the centre (1).	1+1+2+1
20	(i) On the perpendicular bisector of AB (1). (ii) Line drawing (2). (iii) The midpoint (1). (iv) 3 km (1).	1+2+1+1
	OR: (i) Copy (1). (ii) Square (2). (iii) Sides, vertices, diagonals (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.